Hard Coral Cover Data Selection and Exploration

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Data Source and Selection Criteria

We utilized benthic community data from the National Coral Reef Monitoring Program (NCRMP) for St. Thomas and St. John (STTSTJ), U.S. Virgin Islands. The dataset spans from 2013 to 2025 and contains site-level percent cover estimates for eight benthic functional groups: crustose coralline algae (CCA), hard corals, macroalgae, other substrates, soft corals, sponges, turf algae, and bare substrate.

```r
# Define path
path_csv <- "G:/Shared drives/NSF CoPE internal/GIS_CoPE/GIS_USVI/2_model_inputs_usvi/Fisheries/coral_c

# Load and clean names
coral_raw <- read_csv(path_csv) %>%
  clean_names()

# Initial inspection
glimpse(coral_raw)
```

```
## Rows: 13,744
## Columns: 16
## $ region             <chr> "STTSTJ", "STTSTJ", "STTSTJ", "STTSTJ", "STTSTJ", ~
## $ year               <dbl> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 20~
## $ sub_region_name    <chr> "MSR", "MSR", "MSR", "MSR", "MSR", "MSR", "MSR", "~
## $ admin              <chr> "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "O~
## $ primary_sample_unit <dbl> 8947, 8947, 8947, 8947, 8947, 8947, 8947, 8947, 89~
## $ lat_degrees        <dbl> 18.2758, 18.2758, 18.2758, 18.2758, 18.2758, 18.27~
## $ lon_degrees        <dbl> -64.9678, -64.9678, -64.9678, -64.9678, -64.9678, ~
## $ min_depth          <dbl> 27.4320, 27.4320, 27.4320, 27.4320, 27.4320, 27.43~
## $ max_depth          <dbl> 29.2608, 29.2608, 29.2608, 29.2608, 29.2608, 29.26~
## $ analysis_stratum   <chr> "AGRF_DEEP", "AGRF_DEEP", "AGRF_DEEP", "AGRF_DEEP"~
## $ strat              <chr> "AGRF_DEEP", "AGRF_DEEP", "AGRF_DEEP", "AGRF_DEEP"~
## $ habitat_cd         <chr> "AGRF", "AGRF", "AGRF", "AGRF", "AGRF", "AGRF", "A~
## $ prot               <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ cover_group        <chr> "CCA", "HARD CORALS", "MACROALGAE", "OTHER", "SOFT~
## $ percent_cvr        <dbl> 2.000000, 9.000000, 41.000000, 13.000000, 0.000000~
## $ n                  <dbl> 1, 1, 1, 1, 0, 1, 1, 0, 1, 0, 1, 1, 0, 1, 1, 0, 1,~
```

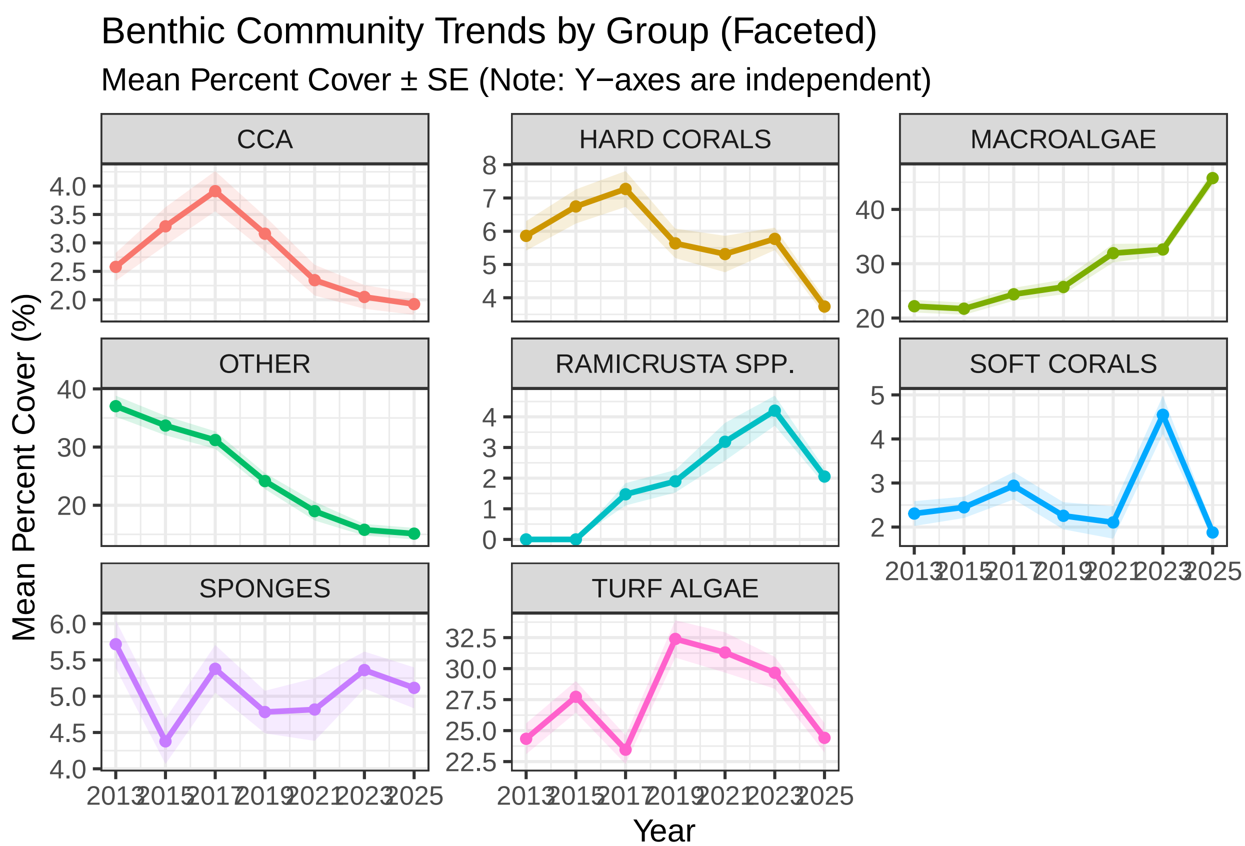
The NCRMP dataset contained 13,744 total observations across all years, regions, and benthic groups. Each observation represents percent cover at a primary sampling unit (PSU), with associated metadata including:

- Spatial information: Latitude and longitude (decimal degrees), depth range
- Stratification variables: Analysis stratum, habitat code, protection status
- Administrative data: Sub-region, management designation

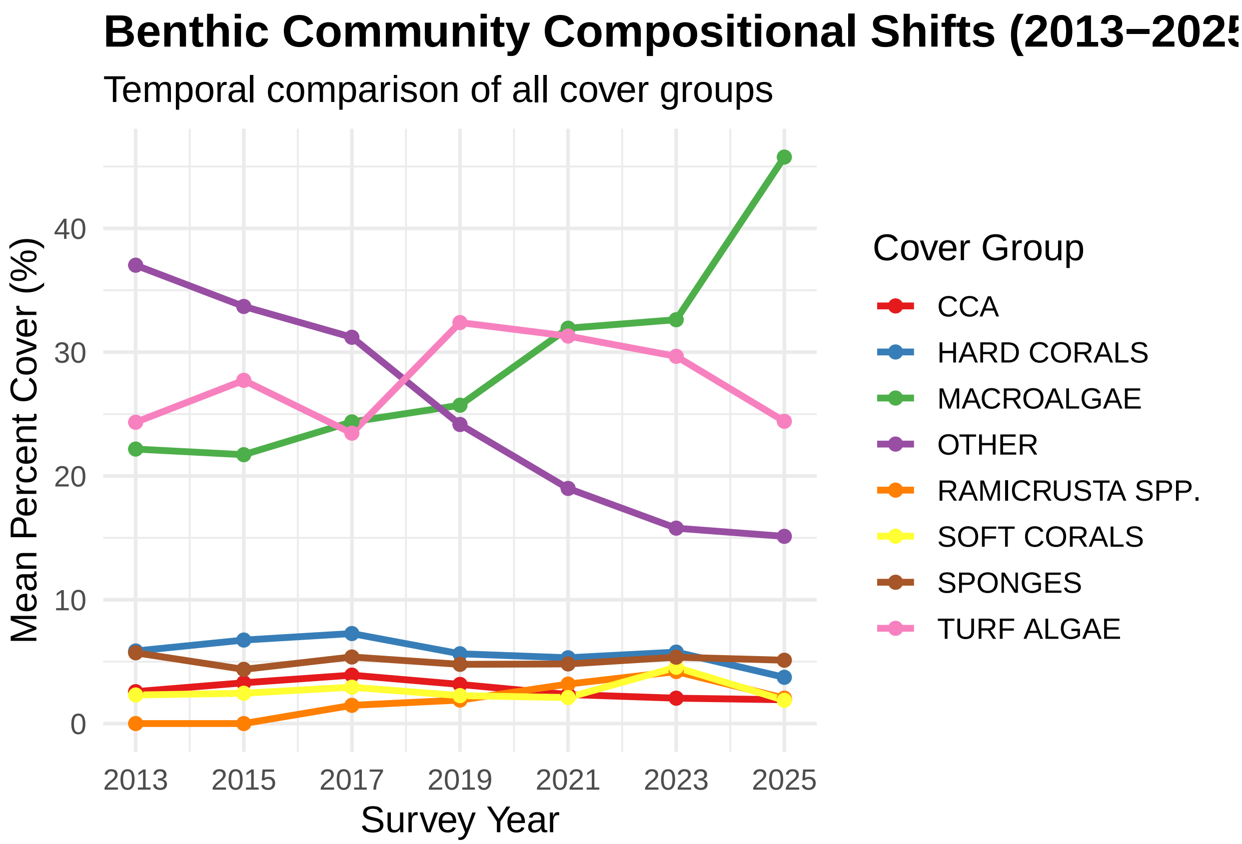
Visualization

```r
## 1. Prepare Data for All Groups
# We calculate mean and standard error for every benthic category
benthic_trends <- coral_raw %>%
  group_by(year, cover_group) %>%
  summarise(
    n = n(),
    mean_cover = mean(percent_cvr, na.rm = TRUE),
    se_cover = sd(percent_cvr, na.rm = TRUE) / sqrt(n),
    .groups = 'drop'
  )

## 2. Faceted Plot (Individual plots for each group)
# This allows for clear inspection of groups with very different scales
ggplot(benthic_trends, aes(x = year, y = mean_cover, color = cover_group)) +
  geom_ribbon(aes(ymin = mean_cover - se_cover, ymax = mean_cover + se_cover, fill = cover_group),
              alpha = 0.15, color = NA) +
  geom_line(size = 1) +
  geom_point() +
  facet_wrap(~cover_group, scales = "free_y") +
  theme_bw(base_size = 12) +
  theme(legend.position = "none") +
  # --- X-Axis Adjustment Starts Here ---
  scale_x_continuous(
    breaks = unique(benthic_trends$year),
    labels = as.character(unique(benthic_trends$year))
  ) +
  # --- X-Axis Adjustment Ends Here ---
  labs(
    title = "Benthic Community Trends by Group (Faceted)",
    subtitle = "Mean Percent Cover ± SE (Note: Y-axes are independent)",
    x = "Year",
    y = "Mean Percent Cover (%)"
  )
```

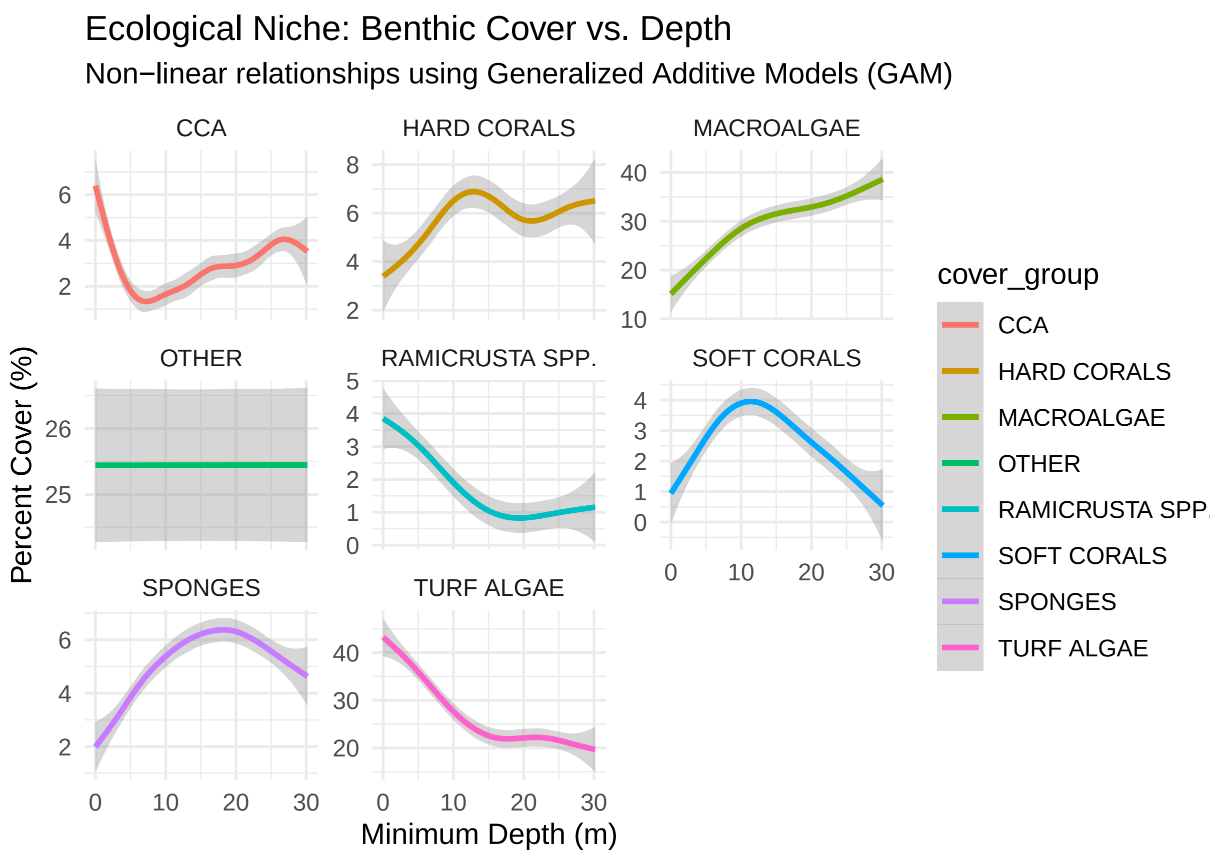



```r
## 3. Overlay Plot (All groups in one professional visualization)
# This shows the relative dominance of different groups over time
ggplot(benthic_trends, aes(x = year, y = mean_cover, color = cover_group, group = cover_group)) +
  geom_line(size = 1.2) +
  geom_point(size = 2) +
  scale_color_brewer(palette = "Set1") + # Distinct colors for the 8 groups
  theme_minimal(base_size = 14) +
  labs(
    title = "Benthic Community Compositional Shifts (2013-2025)",
    subtitle = "Temporal comparison of all cover groups",
    x = "Survey Year",
    y = "Mean Percent Cover (%)",
    color = "Cover Group"
  ) +
  scale_x_continuous(breaks = seq(2013, 2025, by = 2)) +
  theme(
    legend.position = "right",
    plot.title = element_text(face = "bold")
  )
```

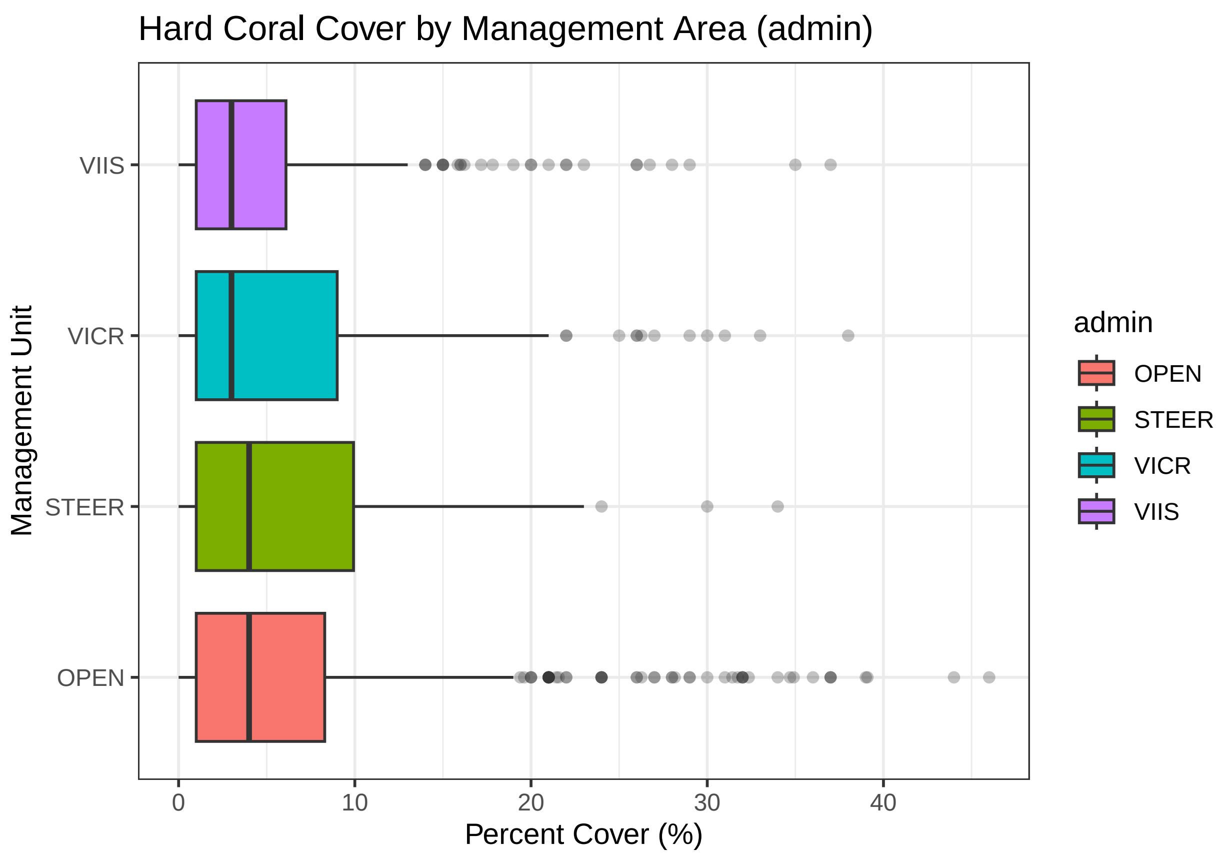


This code investigates the relationship between Depth and Cover Group, and explores how Habitat affects the distribution

```r
## 1. Depth-Response Curves by Group
# This helps identify which groups are depth-sensitive
ggplot(coral_raw, aes(x = min_depth, y = percent_cvr, color = cover_group)) +
  geom_smooth(method = "gam", formula = y ~ s(x, bs = "cs"), se = TRUE) +
  facet_wrap(~cover_group, scales = "free_y") +
  theme_minimal() +
  labs(
    title = "Ecological Niche: Benthic Cover vs. Depth",
    subtitle = "Non-linear relationships using Generalized Additive Models (GAM)",
    x = "Minimum Depth (m)",
    y = "Percent Cover (%)"
  )
```




```r
## 2. Categorical Distribution: Habitat & Management
# Let's look at Hard Corals specifically across different Management Zones (admin)
coral_raw %>%
  filter(cover_group == "HARD CORALS") %>%
  ggplot(aes(x = admin, y = percent_cvr, fill = admin)) +
  geom_boxplot(outlier.alpha = 0.3) +
  theme_bw() +
  coord_flip() + # Better for long labels
  labs(
    title = "Hard Coral Cover by Management Area (admin)",
    x = "Management Unit",
    y = "Percent Cover (%)"
  )
```




```r
## 3. Correlation Matrix (Numeric)
# Checking if max_depth and min_depth are redundant (collinearity check)
library(corrr)
coral_raw %>%
  select(min_depth, max_depth, percent_cvr) %>%
  correlate() %>%
  shave() %>%
  fashion()
```

```
##          term min_depth max_depth percent_cvr
## 1   min_depth
## 2   max_depth       .99
## 3 percent_cvr      -.00      -.00
```

Filter 2025 data

All 276 hard coral observations for 2025 had complete geographic coordinates (100% spatial completeness), ensuring no data loss during the planned spatial join with environmental predictor layers in ArcGIS.

Data Structure and Completeness

For our boosted regression tree (BRT) modeling framework, we selected hard coral cover as a key environmental predictor variable based on its established ecological importance as habitat structure for reef fish assemblages. We filtered the data to include only:

- Year: 2025 (matching the fish biomass survey year)
- Region: STTSTJ (St. Thomas and St. John)
- Benthic group: HARD CORALS

This selection resulted in 276 sampling points with complete geographic coordinates, ready for spatial integration with the 50m × 50m modeling grid.

```r
# Filter for hard corals in 2025 and extract coordinates
hard_coral_2025 <- coral_raw %>%
  filter(
    year == 2025,
    cover_group == "HARD CORALS"
  ) %>%
  select(
    primary_sample_unit,
    region,
    lat = lat_degrees,  # Adjust column name if different
    lon = lon_degrees,  # Adjust column name if different
    hard_coral_cover = percent_cvr,
    max_depth = max_depth,    # Include if available
    everything()        # Keep all other columns for reference
  )

# Summary of coral data
cat("\n=== HARD CORAL COVER DATA SUMMARY (2025) ===\n\n")
```

```
##
## === HARD CORAL COVER DATA SUMMARY (2025) ===
```

```r
cat("Total sampling points:", nrow(hard_coral_2025), "\n")
```

```
## Total sampling points: 276
```

```r
cat("Regions:", paste(unique(hard_coral_2025$region), collapse = ", "), "\n\n")
```

```
## Regions: STTSTJ
```

```r
# Check for missing coordinates
cat("=== COORDINATE COMPLETENESS ===\n")
```

```
## === COORDINATE COMPLETENESS ===
```

```r
cat("Points with complete coordinates:",
    sum(!is.na(hard_coral_2025$lat) & !is.na(hard_coral_2025$lon)), "\n")
```

```
## Points with complete coordinates: 276
```

```r
cat("Points missing lat:", sum(is.na(hard_coral_2025$lat)), "\n")
```

```
## Points missing lat: 0
```

```r
cat("Points missing lon:", sum(is.na(hard_coral_2025$lon)), "\n\n")
```

```
## Points missing lon: 0
```

```r
# Statistical summary of hard coral cover
cat("=== HARD CORAL COVER STATISTICS ===\n\n")
```

```
## === HARD CORAL COVER STATISTICS ===
```

```r
coral_stats <- hard_coral_2025 %>%
  summarise(
    n_sites = n(),
    mean_cover = round(mean(hard_coral_cover, na.rm = TRUE), 2),
    sd_cover = round(sd(hard_coral_cover, na.rm = TRUE), 2),
    median_cover = round(median(hard_coral_cover, na.rm = TRUE), 2),
    min_cover = round(min(hard_coral_cover, na.rm = TRUE), 2),
    max_cover = round(max(hard_coral_cover, na.rm = TRUE), 2),
    q25_cover = round(quantile(hard_coral_cover, 0.25, na.rm = TRUE), 2),
    q75_cover = round(quantile(hard_coral_cover, 0.75, na.rm = TRUE), 2)
  )

print(coral_stats)
```

```
## # A tibble: 1 x 8
##   n_sites mean_cover sd_cover median_cover min_cover max_cover q25_cover
##     <int>      <dbl>    <dbl>        <dbl>     <dbl>     <dbl>     <dbl>
## 1     276       3.74     4.13            3         0        29         0
## # i 1 more variable: q75_cover <dbl>
```

```r
# Summary by region
cat("\n=== HARD CORAL COVER BY REGION ===\n")
```

```
##
## === HARD CORAL COVER BY REGION ===
```

```r
coral_by_region <- hard_coral_2025 %>%
  group_by(region) %>%
  summarise(
    n_sites = n(),
    mean_cover = round(mean(hard_coral_cover, na.rm = TRUE), 2),
    sd_cover = round(sd(hard_coral_cover, na.rm = TRUE), 2),
    median_cover = round(median(hard_coral_cover, na.rm = TRUE), 2),
    min_cover = round(min(hard_coral_cover, na.rm = TRUE), 2),
    max_cover = round(max(hard_coral_cover, na.rm = TRUE), 2),
    .groups = "drop"
  )

print(coral_by_region)
```

```
## # A tibble: 1 x 7
##   region n_sites mean_cover sd_cover median_cover min_cover max_cover
##   <chr>    <int>      <dbl>    <dbl>        <dbl>     <dbl>     <dbl>
## 1 STTSTJ     276       3.74     4.13            3         0        29
```

```r
# Spatial extent check
cat("\n=== SPATIAL EXTENT ===\n")
```

```
##
## === SPATIAL EXTENT ===
```

```r
spatial_extent <- hard_coral_2025 %>%
  summarise(
    min_lat = round(min(lat, na.rm = TRUE), 4),
    max_lat = round(max(lat, na.rm = TRUE), 4),
    min_lon = round(min(lon, na.rm = TRUE), 4),
    max_lon = round(max(lon, na.rm = TRUE), 4)
  )

print(spatial_extent)
```

```
## # A tibble: 1 x 4
##   min_lat max_lat min_lon max_lon
##     <dbl>   <dbl>   <dbl>   <dbl>
## 1    18.2    18.4   -65.1   -64.6
```

```r
# Export for ArcGIS
write.csv(hard_coral_2025,
          "hard_coral_cover_2025_points.csv",
          row.names = FALSE)

cat("\n=== CORAL DATA EXPORTED ===\n")
```

```
##
## === CORAL DATA EXPORTED ===
```

```r
cat("File: hard_coral_cover_2025_points.csv\n")
```

```
## File: hard_coral_cover_2025_points.csv
```

```r
cat("Points:", nrow(hard_coral_2025), "\n")
```

```
## Points: 276
```

```r
cat("Ready for ArcGIS 'Value to Point' tool\n\n")
```

```
## Ready for ArcGIS 'Value to Point' tool
```

```r
cat("Key columns for spatial join:\n")
```

```
## Key columns for spatial join:
```

```r
cat("  - lat: Latitude (decimal degrees)\n")
```

```
##   - lat: Latitude (decimal degrees)
```

```r
cat("  - lon: Longitude (decimal degrees)\n")
```

```
##   - lon: Longitude (decimal degrees)
```

```r
cat("  - hard_coral_cover: Hard coral percent cover\n")
```

```
##   - hard_coral_cover: Hard coral percent cover
```

```r
cat("  - region: Survey region\n\n")
```

```
##   - region: Survey region
```

```r
# Preview
cat("=== FIRST 10 ROWS ===\n")
```

```
## === FIRST 10 ROWS ===
```

```r
hard_coral_2025 %>%
  select(primary_sample_unit, region, lat, lon, hard_coral_cover) %>%
  head(10) %>%
  print()
```

```
## # A tibble: 10 x 5
##    primary_sample_unit region   lat   lon hard_coral_cover
##                  <dbl> <chr>  <dbl> <dbl>            <dbl>
##  1                5000 STTSTJ  18.2 -64.7             1
##  2                5001 STTSTJ  18.2 -64.7             7
##  3                5002 STTSTJ  18.2 -64.7             0
##  4                5003 STTSTJ  18.2 -64.7             4
##  5                5251 STTSTJ  18.2 -64.9             0
##  6                5004 STTSTJ  18.3 -64.9             4
##  7                5005 STTSTJ  18.2 -64.8             4.55
##  8                5006 STTSTJ  18.2 -64.8             4
##  9                5007 STTSTJ  18.3 -64.9             5
## 10                5008 STTSTJ  18.3 -64.8             8
```

Hard Coral Cover Distribution (2025)

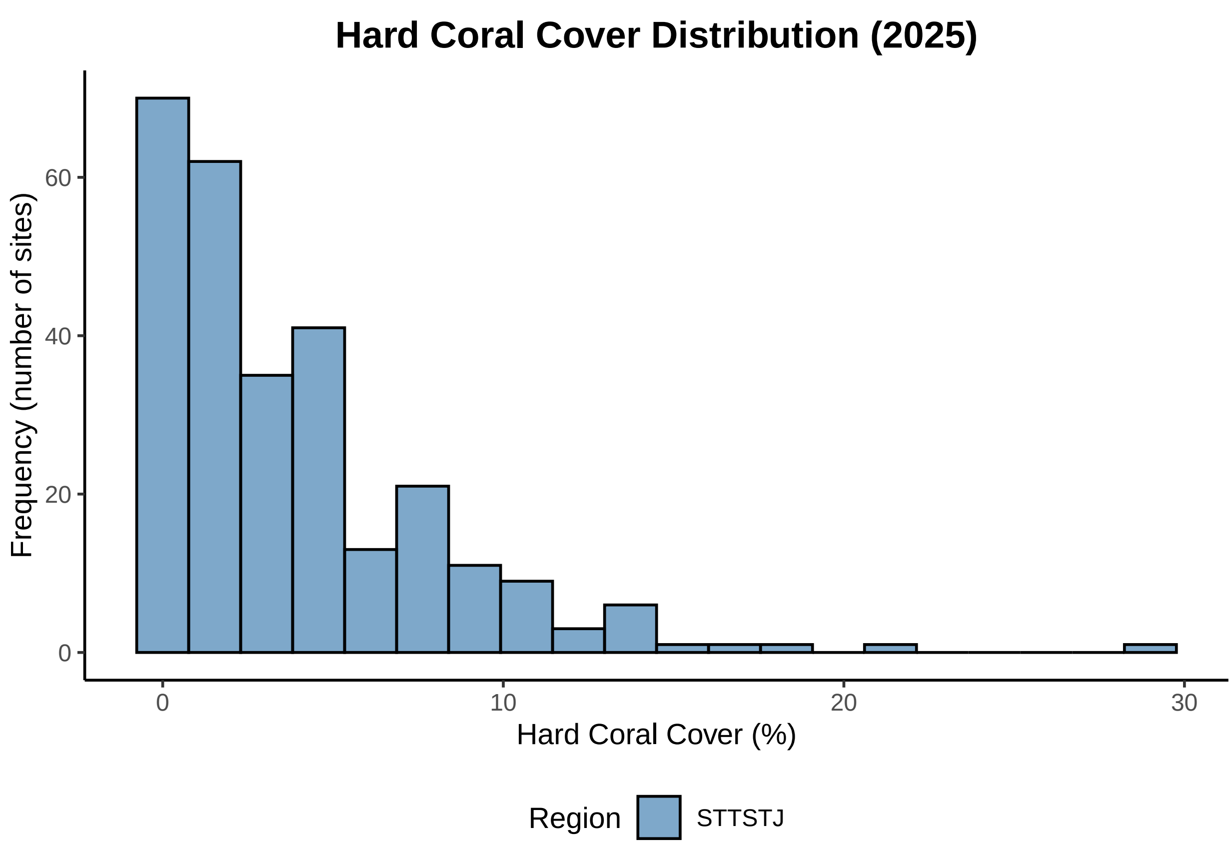
Overall Statistics Hard coral cover across the 276 sampling sites exhibited low mean cover with high spatial heterogeneity:

__ Mean cover: 3.74% (SD = 4.13%) - Median cover: 3.00% - Range: 0–29% - Interquartile range: 0–5% (Q25 = 0%, Q75 = 5%)

The distribution was right-skewed, with 50% of sites having <=3% hard coral cover and 75% of sites having <=5% cover. This indicates a reef system dominated by low coral cover, consistent with regional patterns of coral decline in the Caribbean. The maximum observed cover of 29% was substantially higher than the mean, suggesting a few sites with relatively healthy coral communities persist within the study area.

Hard Coral Cover Visualization

```
# Histogram
ggplot(hard_coral_2025, aes(x = hard_coral_cover, fill = region)) +
  geom_histogram(bins = 20, alpha = 0.7, color = "black") +
  scale_fill_manual(values = c("STTSTJ" = "steelblue", "STX" = "coral")) +
  theme_classic() +
  labs(
    title = "Hard Coral Cover Distribution (2025)",
    x = "Hard Coral Cover (%)",
    y = "Frequency (number of sites)",
    fill = "Region"
  ) +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 14),
    legend.position = "bottom"
  )
```



Spatial Extent

The sampling design covered the insular shelf surrounding St. Thomas and St. John:

- Latitudinal range: 18.1783°N to 18.4139°N (spanning ~26.1 km)
- Longitudinal range: -65.1451°W to -64.6379°W (spanning ~46.2 km)

This spatial coverage encompasses the major reef habitats surrounding both islands, including fringing reefs, patch reefs, and shelf-edge habitats at varying depths.

```
# Spatial map of coral cover with colorblind-friendly palette
ggplot(hard_coral_2025, aes(x = lon, y = lat, color = hard_coral_cover,
                             size = hard_coral_cover)) +
  geom_point(alpha = 0.7) +
  scale_color_gradientn(
    colors = c("#440154", "#3b528b", "#21918c", "#5ec962", "#fde724"),
    name = "Hard Coral\nCover (%)",
    values = scales::rescale(c(0, 5, 10, 15, 29))
  ) +
  scale_size_continuous(range = c(1, 6), name = "Hard Coral\nCover (%)") +
  theme_minimal() +
  labs(
    title = "Spatial Distribution of Hard Coral Cover (2025)",
    x = "Longitude",
    y = "Latitude"
  ) +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 14),
    legend.position = "right"
  )
```